

14 傅里叶级数, Fourier Analysis

知识点回顾:

- 2π 周期函数傅里叶级数的概念.
- 任意周期函数的傅里叶级数; 函数的周期延拓和奇偶延拓.
- 傅里叶级数的收敛性, Parseval 等式.

问题 14.1. 求 2π 周期函数 $f(x) = \begin{cases} x, & -\pi \leq x < 0; \\ 2x, & 0 \leq x \leq \pi \end{cases}$ 的 Fourier 级数, 并求其 Fourier 级数的和函数.

Solution. 先求其 Fourier 系数.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 x dx + \frac{1}{\pi} \int_0^{\pi} 2x dx = \frac{1}{2}\pi.$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 x \cos nx dx + \frac{1}{\pi} \int_0^{\pi} 2x \cos nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 x \cos nx dx + \frac{1}{\pi} \int_0^{\pi} x \frac{1}{n} d(\sin nx) = -\frac{1}{n\pi} \int_0^{\pi} \sin nx dx \\ &= \frac{1}{n^2\pi} \cos nx \Big|_0^{\pi} = \frac{(-1)^n - 1}{n^2\pi}. \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 x \sin nx dx + \frac{1}{\pi} \int_0^{\pi} 2x \sin nx dx \\ &= \frac{3}{\pi} \int_0^{\pi} x \sin nx dx = -\frac{3}{n\pi} \int_0^{\pi} x d(\cos nx) = \frac{3}{n}(-1)^{n+1}. \end{aligned}$$

因此,

$$f(x) \sim \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{-2}{(2n-1)^2\pi} \cos(2n-1)x + \frac{3(-1)^{n+1}}{n} \sin nx \right].$$

显然 f 分段连续且单调, 故其 Fourier 级数的和函数为

$$S(x) = \begin{cases} x, & -\pi < x < 0; \\ 2x, & 0 \leq x < \pi; \\ \pi/2, & x = \pm\pi. \end{cases}$$

问题 14.2. 设 $f(x)$ 是 2π 周期连续函数, 其傅立叶系数为 $a_0, a_n, b_n (n = 1, 2, \dots)$. 求函数

$$F(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) f(t+x) dt$$

的傅立叶系数 A_0, A_n, B_n .

Solution. Step 1. 说明 F 是偶函数.

$$\begin{aligned} F(-x) &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) f(t-x) dt \xrightarrow{\text{(换元 } s=t-x)} \frac{1}{\pi} \int_{-\pi-x}^{\pi-x} f(s+x) f(s) ds \\ &\xrightarrow{\text{(利用周期性)}} \frac{1}{\pi} \int_{-\pi}^{\pi} f(s+x) f(s) ds = F(x). \end{aligned}$$

因此傅立叶系数 $B_n = 0$.

Step 2. 计算 A_0 .

$$\begin{aligned} A_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} F(x) dx = \frac{1}{\pi^2} \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} f(t) f(t+x) dt \right) dx \\ &= \frac{1}{\pi^2} \int_{-\pi}^{\pi} f(t) \left(\int_{-\pi}^{\pi} f(t+x) dx \right) dt. \end{aligned}$$

计算内层积分, 固定 t , 利用 f 的周期性, 有

$$\int_{-\pi}^{\pi} f(t+x) dx = \int_{-\pi+t}^{\pi+t} f(x) dx = \int_{-\pi}^{\pi} f(x) dx = \pi a_0.$$

因此

$$A_0 = \frac{1}{\pi^2} \int_{-\pi}^{\pi} f(t) \pi a_0 dt = a_0^2.$$

Step 3. 计算 A_n .

$$\begin{aligned} A_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} F(x) \cos nx dx = \frac{1}{\pi^2} \int_{-\pi}^{\pi} \left(\int_{-\pi}^{\pi} f(t) f(t+x) dt \right) \cos nx dx \\ &= \frac{1}{\pi^2} \int_{-\pi}^{\pi} f(t) \left(\int_{-\pi}^{\pi} f(t+x) \cos nx dx \right) dt \end{aligned}$$

计算内层积分, 固定 t ,

$$\begin{aligned} \int_{-\pi}^{\pi} f(t+x) \cos nx dx &= \int_{-\pi+t}^{\pi+t} f(x) \cos n(x-t) dx = \int_{-\pi}^{\pi} f(x) \cos n(x-t) dx \\ &= \cos nt \int_{-\pi}^{\pi} f(x) \cos nx dx + \sin nt \int_{-\pi}^{\pi} f(x) \sin nx dx \\ &= \pi(a_n \cos nt + b_n \sin nt). \end{aligned}$$

因此

$$A_n = \frac{1}{\pi^2} \int_{-\pi}^{\pi} f(t) \pi (a_n \cos nt + b_n \sin nt) dt = a_n^2 + b_n^2.$$

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问题 14.3. (1) 借助函数 $f(x) = \begin{cases} 0, & -1 < x \leq 0; \\ x, & 0 \leq x \leq 1, \end{cases}$ 计算级数 $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$.

(2) 借助函数 $g(x) = |x|, (-\pi \leq x \leq \pi)$, 计算级数 $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}$.

Solutions. (1) 将 f 延拓成 2 周期函数, 计算其 Fourier 系数.

$$a_0 = \int_{-1}^1 f(x) dx = \int_0^1 x dx = \frac{1}{2};$$

$$\begin{aligned} a_n &= \int_{-1}^1 f(x) \cos nx dx = \int_0^1 x \cos n\pi x dx \\ &= \frac{1}{n\pi} x \sin n\pi x \Big|_0^1 - \frac{1}{n\pi} \int_0^1 \sin n\pi x dx = \frac{(-1)^n - 1}{(n\pi)^2}; \end{aligned}$$

$$\begin{aligned} b_n &= \int_{-1}^1 f(x) \sin nx dx = \int_0^1 x \sin n\pi x dx \\ &= -\frac{1}{n\pi} x \cos nx \Big|_0^1 + \int_0^1 \cos nx dx = \frac{(-1)^{n-1}}{n\pi}. \end{aligned}$$

因此

$$f(x) \sim \frac{1}{4} + \sum_{n=1}^{\infty} \left[\frac{-2}{((2n-1)\pi)^2} \cos(2n-1)\pi x + \frac{(-1)^{n-1}}{n\pi} \sin n\pi x \right].$$

因 f 分段连续、单调, 并且 f 在 0 处连续, 故

$$\frac{1}{4} - \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = f(0) = 0.$$

因此,

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}.$$

(2) 将 g 延拓成 2π 周期函数, 计算其 Fourier 系数. 因 g 是偶函数, 故 $b_n = 0$.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |x| dx = \frac{\pi}{2} \int_0^{\pi} x dx = \pi,$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos nx dx = \frac{2[(-1)^n - 1]}{n^2\pi}.$$

由 Parseval 恒等式可得,

$$\frac{\pi^2}{2} + \sum_{n=1}^{\infty} \frac{4[(-1)^n - 1]^2}{n^4\pi^2} = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx = \frac{2}{3}\pi^2.$$

从而解得,

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}.$$

问题 14.4 (傅里叶系数的渐近行为). 设 $f(x)$ 是 2π 周期的 k 阶连续可导函数, 证明其傅里叶系数当 n 充分大时, 有渐近行为 $a_n, b_n = O(n^{-k})$.

Sketch of Proof. 注意到, 因 f 是 2π 周期的, 而且 k 阶连续可导, 故 f 的直到 k 阶的各阶导数也是 2π 周期的. 因此不断分部积分, 可得

$$\begin{aligned}\pi a_n &= \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{n} \left(f(x) \sin nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} f'(x) \sin nx \, dx \right) \\ &= -\frac{1}{n} \int_{-\pi}^{\pi} f'(x) \sin nx \, dx = \frac{1}{n^2} \left(f'(x) \cos nx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} f''(x) \cos nx \, dx \right) \\ &= -\frac{1}{n^2} \int_{-\pi}^{\pi} f''(x) \cos nx \, dx = \dots\dots\end{aligned}$$

可以证明

$$|a_n| \leq \frac{1}{n^k \pi} \int_{-\pi}^{\pi} |f^{(k)}(x)| \, dx \leq \frac{C}{n^k}.$$

对于 b_n , 同理.

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问题 14.5. 设 $f \in C^2[-\pi, \pi]$, 证明其 *Fourier* 级数

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

绝对收敛.

问题 14.6. 设 f 为 2π 周期连续函数, 若其所有 *Fourier* 系数均为 0, 即

$$\begin{aligned}\int_{-\pi}^{\pi} f(x) \cos nx \, dx &= 0, \quad n = 0, 1, 2, \dots; \\ \int_{-\pi}^{\pi} f(x) \sin nx \, dx &= 0, \quad n = 1, 2, \dots.\end{aligned}$$

Sketch of Proof. 利用 Parseval 恒等式即可.

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问题 14.7. 设 $f(x)$ 和 $g(x)$ 都是连续 2π 周期函数, 其傅里叶级数分别为

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \quad g(x) \sim \frac{c_0}{2} + \sum_{n=1}^{\infty} (c_n \cos nx + d_n \sin nx).$$

证明:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x)g(x) \, dx = \frac{a_0 c_0}{2} + \sum_{n=1}^{\infty} (a_n c_n + b_n d_n).$$

Sketch of Proof. 对 $f, g, f+g$ 用 Parseval 等式即可.

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